

RAG to RichEs: Bridging RE Automation Pains with RAG Gains: Keynote Abstract

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Abstract—We’ve all felt it: the gap between what stakeholders say, what developers understand, and what actually gets built. Requirements Engineering (RE) has always been a delicate dance—and often, a painful one. Enter Large Language Models (LLMs), and suddenly the dance floor is crowded with auto-generated specifications, hallucinated user stories, and AI-powered enthusiasm. But enthusiasm alone doesn’t bridge RE gaps. This keynote presents our work on Retrieval-Augmented Generation (RAG) as a way to anchor LLMs in real, contextual information—such as domain documents, UI wireframes, or stakeholder updates. Based on evaluations in real-world settings, RAG-based approaches consistently outperform “vanilla” LLMs on RE automation tasks by keeping generation grounded and focused. We will explore how RAG helps address core automation challenges in RE, share what worked (and what didn’t), and reflect on why RAG is more than another AI buzzword. While not a silver bullet, RAG offers a promising path to contextual, traceable, and human-aligned RE automation.

Index Terms—Requirements Engineering, Retrieval-Augmented Generation (RAG), Large Language Models (LLMs)

I. KEYNOTE COVERAGE

From RE Pain Points to RAG Use Cases. We begin by revisiting common RE bottlenecks—such as ambiguous requirement statements, time-consuming manual processes, and stakeholder misalignment. We then examine the rising use of LLMs in RE automation tasks, alongside the gap between automation outcomes and practical RE pain relief. This keynote highlights how RAG can help alleviate specific pain points by enabling context-aware generation, regulatory mapping, and the construction of intelligent virtual assistants.

Lessons from Real-World Projects. Drawing from our applied research across sectors—including postage and handling [1], the space industry (e.g., regulatory traceability in high-assurance systems), the transport industry [2], and education platforms—I will walk through RAG-based solutions developed or evaluated with practitioners. These examples illustrate how RAG supports tasks such as:

- Building text-based RAG models for RE-driven software quality assurance;
- Extending text-based to multi-modal RAG systems;
- Generating context-aware acceptance criteria from UI mockups and textual specifications;
- Evaluating RAG outputs using human-in-the-loop assessments and traceability metrics;

RAG Under the Microscope: What Can Go Wrong. Not all that glitters is RAG. This section focuses on limitations—such as data leakage, over-reliance on retrieval quality, prompt sensitivity, and challenges in evaluating generated artefacts [3]. I will discuss key failure cases, lessons learned, and the guardrails we introduced to ensure reliability and accountability in RAG-based pipelines.

The Road Ahead: RAG as a RE Partner, Not a Replacement. The keynote concludes with an agenda for making RAG a true collaborator for RE professionals—not a replacement. We recognise that both *the requirements space*—shaped by evolving stakeholder needs and regulatory complexity—and the *RAG/LLM technology landscape* are in constant flux. Navigating this dual evolution demands adaptable tooling, responsible practices, and sustained collaboration between RE researchers, practitioners, and AI developers.

II. BIOGRAPHY

Dr Chetan Arora is the Director of Education (Department of Software Systems and Cybersecurity) and a Senior Lecturer in Software Engineering at Monash University, Australia. He is passionate about Requirements Engineering, software quality assurance, and applied AI research. With extensive experience in both industry and academia, his work focuses on responsible AI, trustworthy systems, and compliance in software development—spanning domains such as space systems, healthcare, and legal tech. Dr Arora received his MSc from Technische Universität Kaiserslautern (Germany) and his PhD from the University of Luxembourg (Luxembourg). He subsequently worked in innovation management at SES Satellites (Luxembourg) on several European space projects before moving into Australian academia. Dr Arora has published over 65 scientific papers and serves on the organising and program committees of several international software engineering conferences. More details: <https://www.drchetanarora.com/>.

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